



1	Course Code	BCI2023											
2	Course Name	DATABASE SYSTEMS											
3	Version												
4	Name(s) of Academic Staff	Noraziah Binti Ahmad											
5	Program Level	Bachelor'S Degree											
6	Credit Hours	3											
7	Student Learning Time (SLT)	Face-to-Face	Normal Teaching & Learning				Online Teaching & Learning				Total SLT		
			Non Face-to-Face			Formal Assessment	Online Learning	Online Activities	Online Assessment	Online Self Learning			
			Guided	Non-Guided	Assessment Preparation								
			52	2	39	18	9	0	0	0		0	120
			120 (0%)					0 (0%)					
8	Prerequisite Course												
9	Contact Hours	Face-to-face Session		Total Hours per semester				Hours per Week					
		Lecture		26				2					
		Tutorial		0									
		Laboratory		26				2					
		Supervision		0									
		Online Learning		0									
		Others		0									
10	Course Synopsis	The course emphasizes on the importance of data to an organization and how the data should be managed. Database management system (DBMS) will be viewed as a solution to the problems of file processing system. Aspects of relational database design will be covered in details. This includes database development life cycle, database architecture, data models, and normalization process. Structured Query Language (SQL) will be discussed and empahsised as query language in database management. Students will be given a real life problem to design and develop a database application system. In the later part of the course students will be exposed to the latest developments in database architecture.											
11	Course Outcome	By the end of semester, student should be able to:											
			CO Statements			Domain	Keyword	Level (Bloom	Sub Keyword (if any)				
		CO1	Demonstrate the concepts and principles of database systems (C4, PL01)			COGNITVE	Knowledge	4					
		CO2	Manipulate queries using the syntax of Structure Query Language (SQL) (P4, PL02)			PSYCHMOTOR	Technical Skills	4					
		CO3	Construct innovative solution through the representation of data model using ER and EER Diagrams and normalize database to be implemented in database application system using appropriate DBMS (C4, PL03)			COGNITVE	Critical Thinking & Problem Solving Skills	4					
		CO4	Work in group in order to complete the given assessments in specific time frame (A3,PL05)			AFFECTIVE	Individual & Teamwork	3					
12	Rationale	Category : Core Programme											
13	Transferable Skills	Category : Critical Thinking & Problem Solving Skills Design ER, EER, normalization of the database logical design. Develop database using SQL											
14	Teaching - learning and Assessment Strategy	Teaching: Lecture & lab (Face to face) Assessment: Quiz, midterm test, HOT, mini project & final examination											

15	Assessment Methods	Methods		Weighting	CO1	CO2	CO3	CO4				
		FINAL EXAM		40 %	40							
		HOT		10 %		10						
		MID TERM		20 %	20							
		MINI PROJECT		25 %		10	10	5				
		QUIZ 1		3 %	2.5							
		QUIZ 2		3 %	2.5							
				100%	65	20	10	5				
16	Learning Reference		Author		Title		Edition	Publisher	Year of Publish	Reference Type	Note	
		1	Carlos Coronel and Steven Morris		Database Systems: Design, Implementation, & Management (MindTap Course List)			Cengage Learning	2022	BOOK		
		2	David Lorenz		Building Production-Grade Web Applications with Supabase: A comprehensive guide to database design, security, real-time data, storage, multi-tenancy, and more.		13th Edition	Packt Publishing	2024	BOOK		
		3	Elvis Foster		Database Systems: A Pragmatic Approach		3rd Edition	Auerbach Publications	2022	BOOK		
		4	Mark L. Gillenson		Fundamentals of Database Management Systems		3rd Edition	Wiley	2023	BOOK		
		5	Ramakrishnan and Gehrke		Database Management Systems		Kindle Edition	Kindle Store	2024	BOOK		
		6	Theophilus Edet		Database Fundamentals: Strategies for Efficient Access to Information and Enhanced Decision Making (Mastering Database Man...)			Course Technology Inc	2024	BOOK		